Yatin Dandi

Learning and Computation in High Dimensions

PhD candidate, EPFL · expected 2027
Advisors: Lenka Zdeborová and Florent Krzakala

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Website · Scholar · GitHub
Lausanne, Switzerland

FEATURE LEARNING · DEPTH & HIERARCHY · TRAINING DYNAMICS · SCALING LAWS · SPIN GLASSES

Summary

My research asks which structures make high-dimensional problems statistically or algorithmically hard, and which permit efficient learning or computation. In particular, I develop predictive theories of feature learning and the role of depth in neural networks, together with theories of dynamics, optimization, and statistical-to-computational gaps in spin glasses and random graphs. This work draws on mathematical tools from probability, optimization, learning theory, and statistical physics.

Research Highlights

FEATURE LEARNING

Asymptotics of feature learning in two-layer networks
Derived exact asymptotics of feature learning in two-layer networks for gradient descent on high-dimensional data, characterizing the spectrum of learned features and generalization via random matrix theory. Showed that reusing batches makes targets learnable that fresh-batch analyses classify as hard, motivating a refined classification through symmetries and label transformations.

ICML 2024 Spotlight · AISTATS 2025 Oral · JMLR 2024 · ICML 2024 · AISTATS 2025

DEPTH & HIERARCHY

Depth as sequential dimension reduction
Led work identifying a computational advantage of depth on hierarchical targets: gradient descent learns successive representations by reducing effective dimension. Follow-ups extend the mechanism to compositional targets, random hierarchy models, scaling laws, and low-degree spectral filtering.

NeurIPS 2025 Spotlight · ICML 2026 · COLT 2026 · HiLD @ ICML 2026

SPIN GLASSES

Statics, dynamics, and statistical-to-computational gaps in spin glasses
Developed the dynamical cavity method into a rigorous analytical technique yielding asymptotically exact descriptions of first-order algorithms, and characterized stable local optima and phase transitions in random graphs and spin glasses, probing statistical-to-computational gaps.

COLT 2026 · arXiv 2025 · SLMath highlight

Selected Publications

Full list of publications: Google Scholar

COLT
2026

Rigorous Asymptotics for First-Order Algorithms Through the Dynamical Cavity Method arXiv
Yatin Dandi, David Gamarnik, Francisco Pernice, Lenka Zdeborová

PREPRINT
2026 · Under review

Deep Learning as Neural Low-Degree Filtering: A Spectral Theory of Hierarchical Feature Learning arXiv · code
Yatin Dandi, Matteo Vilucchio, Luca Arnaboldi, Hugo Tabanelli, Florent Krzakala

NEURIPS
2025 · Spotlight

The Computational Advantage of Depth: Learning High-Dimensional Hierarchical Functions with Gradient Descent arXiv
Yatin Dandi, Luca Pesce, Lenka Zdeborová, Florent Krzakala

PREPRINT
2025 · Under review

Sequential Dynamics in Ising Spin Glasses arXiv · SLMath
Yatin Dandi, David Gamarnik, Francisco Pernice, Lenka Zdeborová

AISTATS
2025 · Oral

A Random Matrix Theory Perspective on the Spectrum of Learned Features and Asymptotic Generalization arXiv
Yatin Dandi, Luca Pesce, Hugo Cui, Florent Krzakala, Yue M. Lu, Bruno Loureiro

AISTATS
2025

Fundamental Computational Limits of Weak Learnability in High-Dimensional Multi-Index Models paper
Emanuele Troiani, **Yatin Dandi**, Leonardo Defilippis, Lenka Zdeborová, Bruno Loureiro, Florent Krzakala

ICML
2024

The Benefits of Reusing Batches for Gradient Descent in Two-Layer Networks arXiv
Yatin Dandi, Emanuele Troiani, Luca Arnaboldi, Luca Pesce, Lenka Zdeborová, Florent Krzakala

ICML
2024 · Spotlight

Asymptotics of Feature Learning in Two-Layer Networks after One Gradient Step arXiv
Hugo Cui, Luca Pesce, **Yatin Dandi**, Florent Krzakala, Yue M. Lu, Lenka Zdeborová, Bruno Loureiro

JMLR
2024

How Two-Layer Neural Networks Learn, One (Giant) Step at a Time paper · arXiv
Yatin Dandi, Florent Krzakala, Bruno Loureiro, Luca Pesce, Ludovic Stephan

PNAS
2024

Sampling with Flows, Diffusion, and Autoregressive Neural Networks from a Spin-Glass Perspective paper · arXiv
Davide Ghio, **Yatin Dandi**, Florent Krzakala, Lenka Zdeborová

PREPRINT
2023 · Under review

Maximally-Stable Local Optima in Random Graphs and Spin Glasses: Phase Transitions and Universality arXiv
Yatin Dandi, David Gamarnik, Lenka Zdeborová

Research Experience

SINCE 2022

PhD Researcher

LAUSANNE, SWITZERLAND

EPFL · Advisors: Lenka Zdeborová and Florent Krzakala

- Develop predictive theories of feature learning in shallow and deep networks using high-dimensional probability, random matrix theory, and statistical physics.
- Contribute more broadly to the mathematics of spin glasses and diffusion models, statistical-to-computational gaps, high-dimensional statistics, and random graphs.

2024 – 2025

Visiting Researcher / Program Associate

CAMBRIDGE AND BERKELEY, USA

MIT and Simons Laufer Mathematical Sciences Institute · Host: David Gamarnik

- Undertook an EPFL Doc.Mobility-funded visit at MIT and participated in SLMath's program on Probability and Statistics of Discrete Structures.
- Developed novel techniques for analyzing first-order optimization algorithms in high dimensions and derived exact dynamical equations for Glauber dynamics in the SK model.

SUMMER 2022

Research Intern

MONTREAL, CANADA

Mila · Advisors: Simon Lacoste-Julien and Yoshua Bengio

- Developed a theoretical framework for analyzing convergence of min-max algorithms, including the saddle-point Frank-Wolfe method.

2019 – 2022

Undergraduate Researcher

INDIA AND SWITZERLAND

IIT Kanpur and EPFL · Collaborators: Martin Jaggi, Piyush Rai, Abhishek Kumar

- Developed optimization methods for heterogeneous distributed learning, yielding work on implicit gradient alignment (AAAI 2022) and decentralized mixing (NeurIPS 2022 workshop).
- Developed generalized adversarially learned inference (AAAI 2021); also co-developed a joint image-video generation model (WACV 2020).

Education

2022 – 2027

EPFL

PhD in Computer Science · expected 2027

Advisors: Lenka Zdeborová and Florent Krzakala.

2017 – 2022

Indian Institute of Technology Kanpur

BTech–MTech in Computer Science and Engineering

CPI: 9.6/10; Master's CPI: 10/10. Thesis: *Tackling Heterogeneity in Collaborative Machine Learning*, co-supervised by Martin Jaggi and Piyush Rai.

Recognition, Talks, and Technical Depth

RECOGNITION

EPFL AI Center PhD Fellowship, 2026 · Jane Street Graduate Research Fellowship (Honorable Mention), 2026 · EPFL Doc.Mobility Grant, 2024 · EDIC Fellowship, EPFL, 2022

SELECTED TALKS

HiLD Workshop, Banff, 2026 · Youth in High-Dimensions, Trieste, 2026 · ProbAI Scaling Laws, Warwick, 2026 · Physics of ML, Stanford, 2025 · SLMath Seminar, Berkeley, 2025 · JMM, Seattle, 2025 · Invited talk: Oberwolfach, 2023

TECHNICAL TOOLKIT

Python, C++, PyTorch, JAX · NumPy, SciPy, scikit-learn · Linux, Git, Docker, LaTeX · Numerical linear algebra; distributed and federated optimization

SERVICE

Reviewer for Annals of Statistics, NeurIPS, ICML, ICLR, AISTATS, TMLR, and SIAM Journal on Mathematics of Data Science. Supervised several master's-level research projects at EPFL.